STUDENT TOOLKIT

EXPERIMENTAL DESIGN PLANNER

A well-designed experiment changes one factor while holding everything else steady. That is how you can trust that your results came from the thing you tested, and not from chance.

INDEPENDENT VARIABLE	DEPENDENT VARIABLE	CONTROLLED VARIABLES
The one thing you change on purpose	What you measure to see the effect	Everything you keep the same

MY CONTROL GROUP

The version that receives no change. It gives you a baseline, so you can tell whether your independent variable made any difference at all.

SAMPLE SIZE AND TRIALS

Run at least three to five trials. A single odd result has less power to mislead you.

Number of trials I will run: ______________________

MATERIALS I NEED

MY PROCEDURE, STEP BY STEP

DESIGN CHECKLIST

[ ] Change only one variable   [ ] Measure precisely   [ ] Repeat your test several times   [ ] Document everything

If your project involves people, animals, or anything hazardous, plan approval into your timeline before you experiment.